

WASP-35b

R-1649

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-35_160109 · created 2026-08-02T08:00:01+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457396.80472 +/- 0.0006 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.131 +/- 0.047
Transit depth (Rp/Rs)^2	1.73 +/- 1.24 [%]
Orbital Inclination (inc)	85.42 +/- 2.04
Airmass coefficient 1 (a1)	1.46 +/- 1.11
Airmass coefficient 2 (a2)	-0.17 +/- 0.13
Scatter in the residuals of the lightcurve fit is	79.58 %
Best Comparison Star	#2 - [351, 289]
Optimal Aperture	0.75
Optimal Annulus	6.75
Transit Duration (day)	0.101 +/- 0.037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.131 ± 0.047 vs 0.1241 (deviation 0.09σ)
Duration fit vs published	0.101 d vs 0.1317 d (deviation 0.48σ)
Mid-time O–C	-0.7 min
Depth SNR	1.6
Residual scatter	79.58 %

Light curve

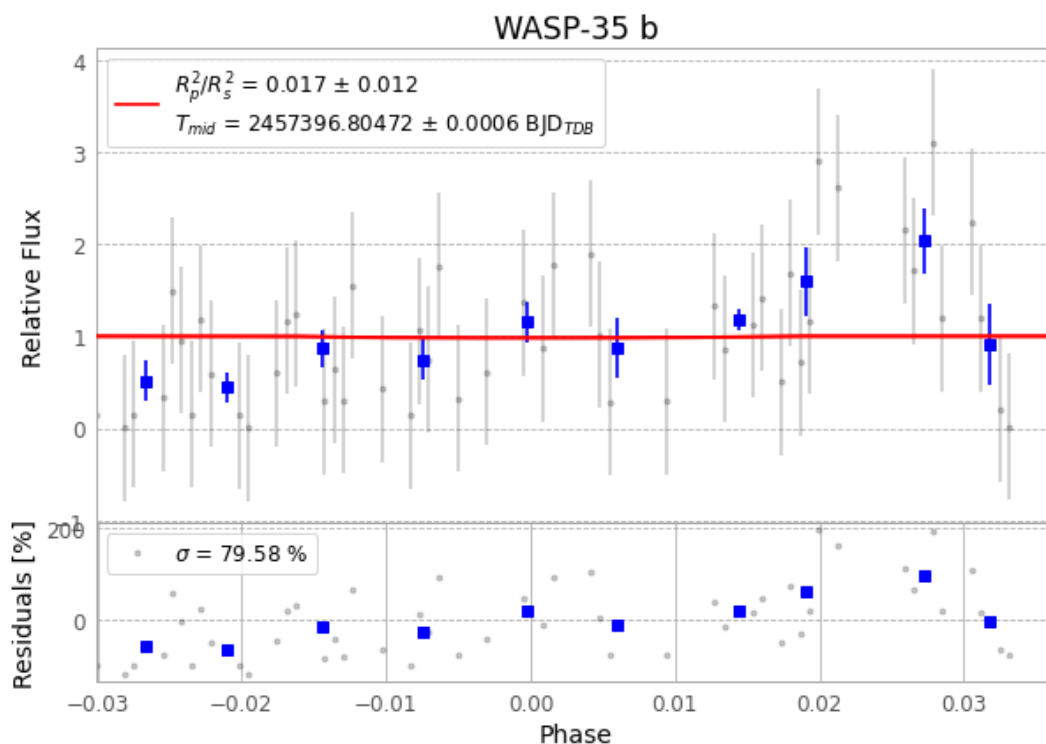


Figure 1 — FinalLightCurve_WASP-35 b_2016-01-08.png (SHA-256 ac6885641c3094d6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [176, 250], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5cb61097c9994165...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

102 FITS frames (67 MB), WASP-35160109045412.FITS → WASP-35160109095712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-160109.log (fb547bb4544d...), FinalLightCurve_WASP-35 b_2016-01-08.png (ac6885641c30...), FinalLightCurve_WASP-35 b_2016-01-08.csv (2829c96d89ef...)

Compute resources

Wall time 270.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 08:00Z.